BLANCA MARTIN-BURGOS

Neural Data Scientist
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EDUCATION

PhD in Neurosciences — UC San Diego, La Jolla, CA | 2020–Present
Computational Neuroscience Specialization
Expected defense: Winter 2026-2027
Advisors: Bradley Voytek and Alysson Muotri

B.A., Neuroscience & Computer Science — Smith College, Northampton, MA | 2016–2020
Highest Honors
Undergraduate thesis: *Computational Analysis of in vivo Circadian Bioluminescence in Freely Moving Mice*

GRADUATE RESEARCH

University of California San Diego | 2020–Present
Voytek Lab & Muotri Lab | Advisors: Bradley Voytek and Alysson Muotri

Action potential waveform variability and neural state
- Developed SpikeWise, an open-source framework for parameterizing intracellular and extracellular action potentials into physiologically interpretable waveform features.
- Analyzed high-resolution intracellular recordings during controlled current injection and simultaneous intracellular and Neuropixels recordings in vivo, showing that within-neuron AP waveform variability systematically reflects neuronal input and local network state.
- Quantified relationships between AP waveform features, subsequent spike timing, recent input statistics, and peri-spike LFP dynamics.
- Work resulted in one submitted manuscript, available as a preprint, and one additional manuscript in preparation.

Neuronal timescales across development
- Studied intrinsic neuronal timescales using spectral methods that separate aperiodic dynamics from oscillatory activity.
- Characterized developmental trajectories of neuronal timescales across human cortical organoids and rat hippocampal cultures.
- Work resulted in a first-author peer-reviewed publication; contributed to an additional methods manuscript in development.

Human cortical organoid electrophysiology
- Developed analysis pipelines to quantify spiking, bursting, and network activity across MEA experiments and stimulation paradigms.
- Implemented and optimized MEA-based electrophysiology approaches for human cortical organoid-derived cultures, including HD-MEA, 3D-MEA, other MEA systems, and electrical and optogenetic stimulation to induce plasticity.
- Contributed experimental and computational analyses across collaborative projects in the Muotri lab, resulting in two co-authored publications and two manuscripts in preparation.

UNDERGRADUATE RESEARCH

Smith College — Harrington Lab | Advisor: Mary Harrington | 2017–2020
- Developed a noninvasive method for recording circadian PER2::LUC bioluminescence rhythms in freely moving mice, combining experimental and computational approaches.
- Developed DWT-based rhythmicity measures and GAM-based waveform comparisons for noisy circadian time series; contributed to CIRCADA educational tools for circadian time-series analysis.
- Work resulted in one first-author and three co-authored peer-reviewed publications.

MIT — Neuroscience Statistics Research Lab | Advisor: Emery Brown | Summer 2019
- Applied point-process modeling to multivariate neural spiking data, focusing on simultaneous events across recorded neuronal populations.

PUBLICATIONS

Martin-Burgos, B., Juavinett, A. L., Riviere, P. D., Hammonds, R., & Voytek, B. (2026). Action potential waveforms are state-dependent. *bioRxiv*. Under review at *Nature Neuroscience*.

LaMontagne, E., Savchenko, A., Gonzalez, G., Vatsyayan, R., **Martin-Burgos, B.**, Puppo, F., et al. (2025). Graphene-polymer nanofibers enable optically induced electrical responses in stem cell-derived electrically excitable cells and brain organoids. *Biomaterials*, *323*, 123430.

Martin-Burgos, B., McPherson, T. S., Hammonds, R., Gao, R., Muotri, A. R., & Voytek, B. (2024). Development of neuronal timescales in human cortical organoids and rat hippocampus dissociated cultures. *Journal of Neurophysiology*, *132(3)*, 757–764.

Dixon, T., **Martin-Burgos, B.**, Puppo, F., Chu, T., & Muotri, A. R. (2023). Modeling ketamine effects on in vitro 3D human neuronal cultures. *Biological Psychiatry*, *93(9 Suppl.)*, S117–S118.

Martin-Burgos, B., Wang, W., William, I., Tir, S., Mohammad, I., Javed, R., et al. (2022). Methods for detecting PER2::LUCIFERASE bioluminescence rhythms in freely moving mice. *Journal of Biological Rhythms*, *37(1)*, 78–93.

Smith, C. B., van der Vinne, V., McCartney, E., Stowie, A. C., Leise, T. L., **Martin-Burgos, B.**, et al. (2022). Cell-type-specific circadian bioluminescence rhythms in Dbp reporter mice. *Journal of Biological Rhythms*, *37(1)*, 53–77.

van der Vinne, V., **Martin-Burgos, B.**, Harrington, M. E., & Weaver, D. R. (2020). Deconstructing circadian disruption: Assessing the contribution of reduced peripheral oscillator amplitude on obesity and glucose intolerance in mice. *Journal of Pineal Research*, *69(1)*, e12654.

Cenek, L., Klindziuk, L., Lopez, C., McCartney, E., **Martin-Burgos, B.**, Tir, S., Harrington, M. E., & Leise, T. L. (2020). CIRCADA: Shiny apps for exploration of experimental and synthetic circadian time series with an educational emphasis. *Journal of Biological Rhythms*, *35(2)*, 214–222.

FELLOWSHIPS, GRANTS & AWARDS

- Graduate Scholarly Impact Award, UC San Diego (Fall 2026)
- CIRM Stem Cell Biology Predoctoral Training Fellowship (2023–2025) — stipend, full tuition, and research/travel support for the duration of the fellowship.
- Washington University in St. Louis Spencer T. & Ann W. Olin Fellowship for Women in Graduate Study (2020; awarded, declined) — stipend, full tuition remission, and travel support for the duration of the fellowship.
- Neuroscience Department Highest Honors (2020)
- McKinley Honors Research Fellowship (2019–2020)
- Smith College Summer Undergraduate Research Fellowship (2018, 2020; awarded twice)
- Nancy Kershaw Tomlinson Memorial Research Fund (2019)
- AEMES Conference Funding Award (2019)
- Schultz Foundation Conference Travel Award (2018, 2019)

TEACHING & MENTORSHIP

Teaching Assistant, Introduction to Python for Biologists (BILD 62) — UC San Diego

- Led four weekly discussion sections serving 100+ students, teaching Python programming and data analysis (NumPy, pandas, data visualization) for the life sciences; developed course assignments, graded coursework, and held office hours.

Research Mentorship — UC San Diego and Smith College

- Mentored undergraduate and master's students in neural data analysis, electrophysiology methods, and research best practices.
- Mentee received honorable mention for the CRA (Computing Research Association) Undergraduate Research Award (2025).

INVITED TALKS & WORKSHOPS

Building AI-Aware Assignments for Neuroscience Education. Professional Development Workshop, Society for Neuroscience (SfN), Washington, D.C. (November 2026). Roles: Mary Harrington (Lead Organizer/Moderator); Blanca Martin-Burgos, Ashley L. Juavinett, and T. Tan (Speakers).

A hands-on technical workshop for cognitive neuroscientists. Cognitive Neuroscience Society (CNS), Toronto, Canada (April 2024). Role: Workshop Assistant.

SELECTED COURSEWORK

Deep Learning for Neuroscientists · Computational Machine Learning · Artificial Intelligence · Neurodynamics · Biophysics of Neurons & Networks · Probability & Statistics · Linear Algebra · Algorithms

TECHNICAL SKILLS

Signal Processing & Computational Analysis
Spectral and time–frequency analysis, filtering, wavelets, point-process and multivariate time-series modeling; machine learning, statistical modeling, hypothesis testing, model validation

Programming & Data Analysis
Python (NumPy, SciPy, pandas, scikit-learn, MNE), MATLAB, R; Git/GitHub; reproducible analysis pipelines; AI-assisted scientific workflows and LLM prompting

Experimental & Wet-Lab Methods
Electrophysiology (MEA/HD-MEA, multielectrode probes), calcium imaging, tissue culture, rodent surgeries and behavioral assays

LANGUAGES

Spanish: Native · English: Bilingual (C2) · French: Advanced (C1) · German: Beginner (B1)

CONFERENCE POSTERS

Martin-Burgos, B., Riviere, P. D., Hammonds, R., Juavinett, A. L., & Voytek, B. (2025, November). *Beyond All-or-Nothing: Action Potentials Dynamically Vary with Stimulation and Network State*. Society for Neuroscience (SfN), San Diego, CA.

Martin-Burgos, B., Riviere, P. D., Hammonds, R., Juavinett, A. L., & Voytek, B. (2024, October). *Not all-or-nothing: Intracellular action potential waveform varies systematically with causal stimulation and extracellular oscillatory state*. Society for Neuroscience (SfN), Chicago, IL.

Martin-Burgos, B., Riviere, P. D., Hammonds, R., Juavinett, A. L., & Voytek, B. (2024, June). *Not all-or-nothing: Intracellular action potential waveform varies systematically with extracellular gamma oscillatory state*. Federation of European Neuroscience Societies (FENS), Vienna, Austria.

Martin-Burgos, B., Riviere, P. D., Hammonds, R., & Voytek, B. (2024, April). *Not all-or-nothing: Intracellular action potential waveform varies systematically with extracellular gamma oscillatory state*. Cognitive Neuroscience Society (CNS), Toronto, Canada.

Martin-Burgos, B., Puppo, F., Ganapathy, S., Hammonds, R., Riviere, P. D., Muotri, A. R., & Voytek, B. (2023, June). *Parameterization of neuronal action potential waveform in high-density MEA and patch clamp recordings of control and autistic-patient derived human cortical organoids*. International Society for Stem Cell Research (ISSCR), Boston, MA.

Martin-Burgos, B., Chu, T., McPherson, T. S., Hammonds, R., Puppo, F., Muotri, A. R., & Voytek, B. (2022, November). *Neuronal timescales across the lifespan: Using human cortical organoids to study neurodevelopment and aging*. Society for Neuroscience (SfN), San Diego, CA.

Martin-Burgos, B., McPherson, T. S., Hammonds, R., & Voytek, B. (2022, March). *Nonlinear development of neuronal timescales in human cortical organoids and rat hippocampus dissociated cultures*. Cognitive Neuroscience Society (CNS), San Francisco, CA.

Martin-Burgos, B., McCartney, E., Weaver, D. R., & Harrington, M. E. (2020, June). *Effects of food restriction on circadian rhythms of gene expression recorded in vivo using DBP reporter mice*. Society for Research on Biological Rhythms (SRBR), Virtual.

Martin-Burgos, B., Molyneux, P. C., van der Vinne, V., Leise, T. L., Weaver, D. R., & Harrington, M. E. (2019, October). *Measuring circadian bioluminescence from mice lacking a functional central clock*. Society for Neuroscience (SfN), Chicago, IL.

Wang, W., William, I., Smith, S., **Martin-Burgos, B.**, Tir, S., Javed, R., Molyneux, P. C., Leise, T. L., Takahashi, J. S., Miller, S. C., & Harrington, M. E. (2018, May). *Measuring circadian bioluminescence from freely behaving mice*. Society for Research on Biological Rhythms (SRBR), Jacksonville, FL.